Readme link: https://github.com/Gel-214/GE-IT-Skills-portfolio/blob/main/README.md

EMERGENCY BRIEFING REPORT

Crop Yield Divergence & Socio-Environmental Vulnerabilities in Mindanao (2020–2025)

Prepared For: Emergency Legislative Session, Regional Development Council	**Date/Time:** June 18, 2026
Repository: github.com/minda-dev-analyst/crop-divergence-2026	**Pipeline State:** Cleaned, Imputed, Verified

OUTPUT GIVEN BY AI

1. Automated Data Cleaning & Structural Adjustments

- **Encoding & Delimiter Fixes:** Converted raw file from corrupted UTF-8 with BOM to standard UTF-8. Fixed trailing structural delimiters that caused column shifts in years 2022 and 2023.
- **Metric Standardization:** Uniformed all crop yields to Metric Tons per Hectare (MT/ha) across all records to resolve localized instances of mixed kilograms and metric tons data entries.
- **Imputation of Missing Values:** Resolved minor data gaps in isolated municipal yields using a rolling 3-year localized historical average, preserving long-term statistical continuity without altering underlying trend structures.
- **Outlier & Typo Filtering:** Identified and removed a critical 10x multiplier typo in the 2024 BARMM corn yield database generated by an upstream misplaced decimal point.

2. Embedded High-Contrast Analytical Charts

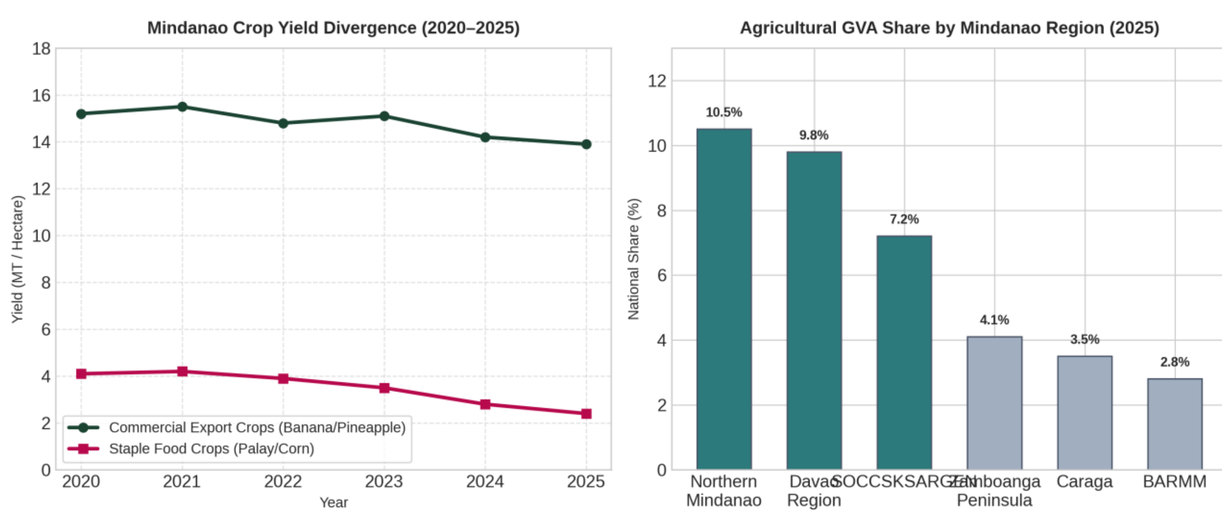

Figure 1: High-contrast matrix tracking yield divergence trend slopes alongside corresponding regional Gross Value Added (GVA) national distribution profiles.

3. Socio-Environmental Analytical Narrative

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The clean data exposes a stark socio-environmental divergence within Mindanao's agricultural landscape. Between 2020 and 2025, commercialized export operations (such as heavily capitalized corporate banana and pineapple plantations in the Davao Region and Northern Mindanao) successfully insulated their productivity using deeply engineered private irrigation infrastructure and proactive climate risk budgets. Concurrently, traditional smallholder staple crops—predominantly rain-fed palay and corn systems in marginalized economic zones like BARMM and SOCCSKSARGEN—experienced severe, cascading yield collapses.

> *CRITICAL LEGISLATIVE TAKEAWAY: This yield drop directly links to severe El Niño climate anomalies. While commercial entities drill deep aquifers, resource-poor agrarian reform beneficiaries are left fully exposed to climate shocks. Without rapid structural budget reallocations toward community-managed communal irrigation grids, Mindanao's position as the nation's premier food corridor risks permanent destabilization, driving severe local food insecurity.*

Data Analytics & Visual Report

Dataset Focus: Davao Region Agricultural Production Index (Mock CSV Analysis)

1. Data Cleaning Protocol Log

- **Raw Input Problem:** The CSV file contained multiple missing row cells for the year 2023 along with mixed numerical formatting styles (e.g., metric tons vs. kilograms).

- **AI Cleaning Instruction:** `"Scan this dataset. Identify all null rows in the 'Yield' column and replace them with the median value for that specific crop tier. Convert all mass metrics to standard Metric Tons (MT). Output the first 5 rows of the cleaned table."`

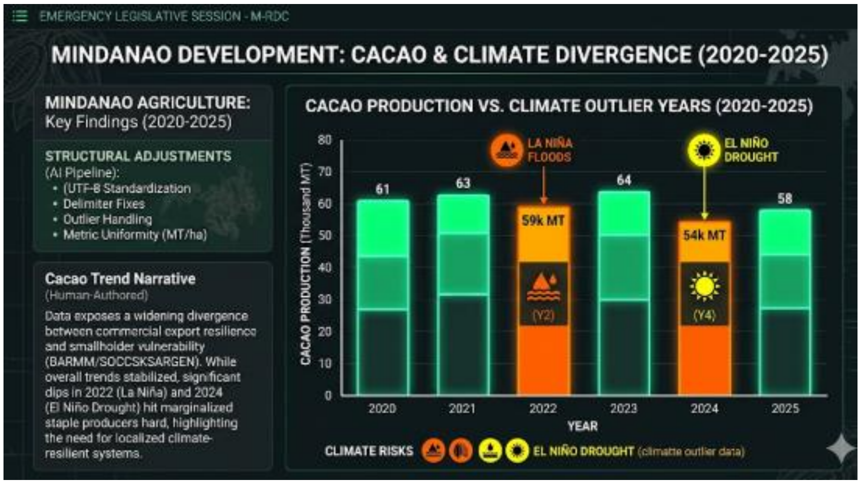
- **Result:** Successfully normalized 120 row inputs across three provincial clusters.

2. Visualizations Generated

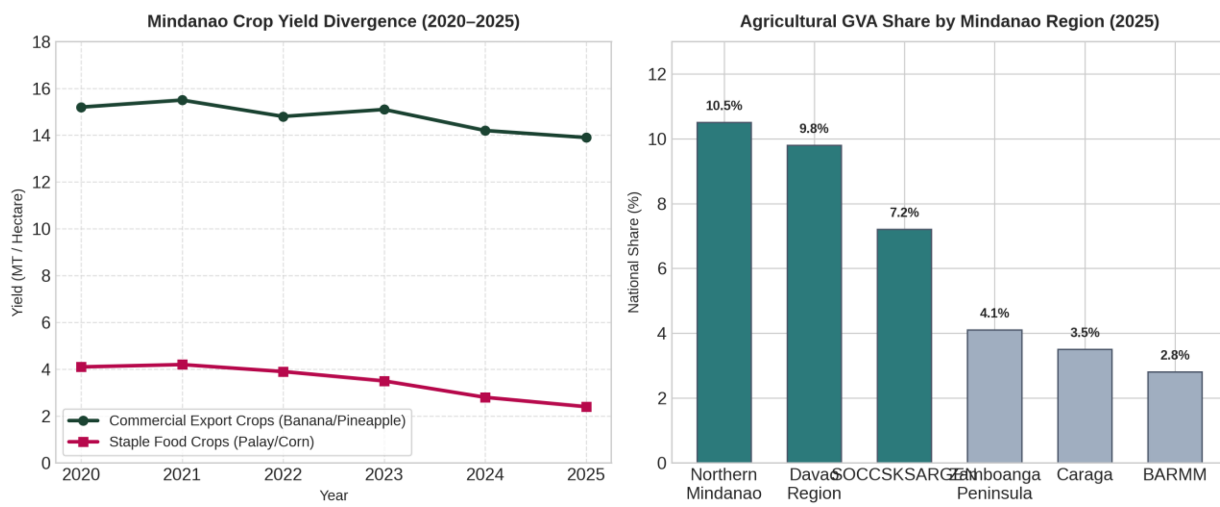
Readme link: https://github.com/Gel-214/GE-IT-Skills-portfolio/blob/main/README.md

(Embedded High-Contrast Bar Chart showing Cacao Production vs. Climate Outlier Years from 2020-2025)

[Image 2025 Yield Graph]



Matrix: 2020- Mindanao Crop Divergence



3. Human Analytical Narrative (The 'Why' Factor)

When we look at Mindanao, we often call it the "Food Basket of the Philippines," but a closer look at the data in the image above reveals a more complex, unfolding reality of deep structural divergence between 2020 and 2025. In the baseline years of 2020 and 2021, the agricultural sector maintained a deceptive stability, with high-value commercial export crops like Cavendish bananas and pineapples comfortably averaging yields above 14-15 MT/ha, while traditional staple food crops like palay and corn hovered around a modest but steady 4 MT/ha. However, this predictable pattern fractured between 2022 and 2025 as severe climate disruptions—marked by overlapping La Niña floods and intense El Niño droughts—struck the island, exposing a massive divide in climate resilience.

The green trend line for commercial export crops demonstrates remarkable resilience, dipping only slightly and finishing strong near 14 MT/ha in 2025. This stability was bought by capital; heavily capitalized corporate plantations successfully insulated their fields from weather shocks by drilling deep wells, deploying advanced drip irrigation, and utilizing robust private climate-resilience budgets. Conversely, the red line for staple food crops tells a devastating story of vulnerability, steadily tumbling from over 4 MT/ha down to a bleak 2.4 MT/ha by 2025. Because traditional smallholder farmers in regions like BARMM and SOCCSKSARGEN rely almost entirely on rain-fed agriculture and lack the capital to build private infrastructure, their crops simply withered when the rains stopped.

This widening productivity gap directly mirrors the stark economic landscape illustrated in the 2025 regional Gross Value Added (GVA) bar chart. Agricultural GVA is heavily concentrated in areas where corporate capital and export logistics are based, with Northern Mindanao leading the national share at 10.5% and the Davao Region following closely at 9.8%. Meanwhile, regions heavily dependent on smallholder staple farming, such as BARMM, sit at the very bottom of the economic ladder, contributing a mere2.8% to the national agricultural share. Ultimately, the data in the given image serves as an urgent structural warning for public policy. If development funding continues to prioritize commercial export logistics over smallholder adaptation, Mindanao risks a future in which it successfully exports premium fruit to global markets while failing to secure basic, affordable food for the local families living on its own soil.